ECE 310 Fall 2026
Lecture 20: Discrete Fourier Transform

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Learning Objectives

After this lecture, you should be able to:

- Explain why the DTFT summation is intractable to fully compute, i.e. in computer code/memory, in practice and why this motivates using the DFT.
- Explain the relationship between the DTFT and the DFT, and describe how the two transforms are different.



How can we compute $X_d(\omega)$ in practice?

For a discrete-time signal $x[n]$, consider computing the DTFT $X_d(\omega)$.

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

More specifically, imagine writing a function in your favorite programming language that returns the DTFT of a given input signal.



How can we compute $X_d(\omega)$ in practice?

In general, we cannot calculate an infinite-length summation in code!



How can we compute $X_d(\omega)$ in practice?

Even if $x[n]$ is finite-length, we cannot evaluate infinitely-many frequencies between $[-\pi, \pi]$!



Discrete Fourier transform (DFT)

The discrete Fourier transform of a length–N discrete-time signal is denoted by $X[k]$.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n},\ 0 \le k \le N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}n},\ 0 \le n \le N-1$$

- Note that $X[k]$ is defined for $0 \le k \le N-1$!



Relationship between DTFT and DFT

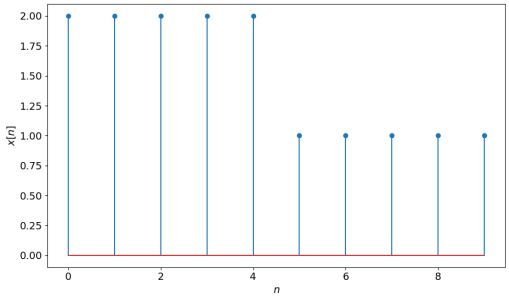
Let $x[n]$ be a length–N discrete-time signal.

$$X_d(\omega) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n}$$

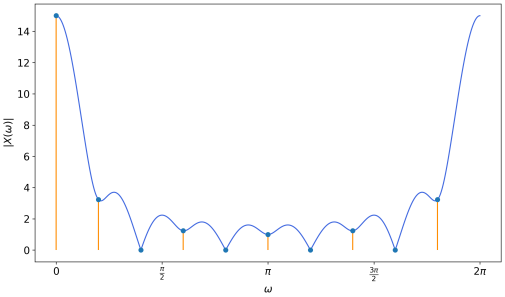
$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N} n}$$



Relationship between DTFT and DFT



(a) Finite-length $x[n]$



(b) $X_d(\omega)$ and values of $X[k]$

N-periodicity of the DFT

For a length–N $x[n] \overset{\text{DFT}}{\longleftrightarrow} X[k]$:

$$X[k] = X[k + mN]$$



N-periodicity of the DFT

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$$X[k] = X[k + mN]$$

Proof:

$$\begin{aligned} X[k+mN] &= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi(k+mN)}{N}n} \\ &= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n} e^{-j2\pi mn} \\ &= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n} \\ &= X[k] \end{aligned}$$



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$$x[n] = x[k + mN]$$

Proof:

$$\begin{aligned} x[n + mN] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}(n+mN)} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}n} e^{j2\pi km} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}n} \\ &= x[n] \end{aligned}$$



How can we get more points from $X_d(\omega)$ in $X[k]$?

What if we want more than N evenly-spaced points from $X_d(\omega)$ evaluated in $X[k]$?



How can we get more points from $X_d(\omega)$ in $X[k]$?

What if we want more than N evenly-spaced points from $X_d(\omega)$ evaluated in $X[k]$?
Let $x[n] \overset{\text{DFT}}{\longleftrightarrow} X[k]$ be a length–N signal. Now, we add L zeros to $x[n]$ to obtain $y[n]$.

$$y[n] = \begin{cases} x[n], & 0 \leq n \leq N-1 \\ 0, & N \leq n \leq N+L-1 \end{cases}$$

What is $Y_d(\omega)$ and $Y[k]$?



Zero-padding Example 1

Let $x[n]$ be a length–16 signal with DFT $X[k]$. We zero-pad $x[n]$ with 32 zeros to obtain $y[n]$ with DFT $Y[k]$. Which values of $Y[k]$ will be equal to:

(a) $X[0]$

(b) $X[1]$

(c) $X[4]$



Zero-padding Example 2

Suppose we have a length–N discrete-time signal $x[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega)$ and we would like the DFT $X[k]$ to contain $X_d\left(\frac{\pi}{5}\right)$ and $X_d\left(\frac{\pi}{3}\right)$. What are the two smallest possible values of N?

